

GSMA Africa Ignite Hackathon 2026 – Idea Phase Submission

Idea Name: SafeTransit AI – Intelligent & Secure Mobility System

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Submitter Details

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Idea Summary

Problem:

Urban mobility in African cities faces significant challenges:

- Lack of real-time vehicle tracking for buses and cabs
- No way for passengers to verify driver identity
- Rising risk of fraud, including SIM swap and device impersonation
- Poor route communication and unpredictable arrival times

These issues reduce public transport adoption, create safety risks, and increase inefficiency in urban mobility.

Proposed Solution:

SafeTransit AI is a telecom-powered, real-time mobility platform designed to improve safety, efficiency, and reliability.

The system will:

- Provide live vehicle tracking using Location Retrieval API
- Authenticate drivers using Number Verification API
- Detect SIM Swap and Device Swap fraud

- Trigger geofencing alerts for arrivals, departures, and route events - Monitor device reachability and network connectivity
- Apply AI-driven predictive ETA and fraud risk scoring for smarter alerts

Expected Benefits:

- **Passenger Safety:** Verified drivers and fraud detection reduce risk
- **Increased Trust:** Reliable tracking and alerts improve confidence in transport
- **Operational Efficiency:** Real-time monitoring reduces waiting times
- **Scalability:** Adaptable across multiple transport modes and cities
- **Digital Inclusion:** Enables smarter, safer mobility for underserved communities

Project Type & GSMA Pillar Alignment

Project Type: Mobile Application (Flutter) and web with Firebase backend integrated with Nokia Network-as-Code APIs

GSMA Pillars: - Digital Identity & Security - Smart Mobility - Network-as-Code Innovation - Digital Inclusion

Theme

Mobility, Transport & Urban Services

API Usage

- **SIM Swap API** – Detects fraudulent SIM changes
- **Device Swap API** – Identifies suspicious device changes
- **Number Verification API** – Authenticates driver phone numbers
- **Location Retrieval API** – Real-time vehicle tracking
- **Geofencing API** – Automated alerts for arrivals/departures
- **Location Verification API** – Confirms device location accuracy
- **Device Status / Reachability API** – Monitors driver connectivity

- **Congestion Insights API** – Improves routing and system reliability
- **Predictive ETA System** – Uses historical + real-time data to estimate arrivals
- **Fraud Risk Scoring Engine** – Combines SIM Swap, Device Swap, and behavior analytics
- **Smart Alerts Engine** – Automatically notifies users of delays, risks, or anomalies